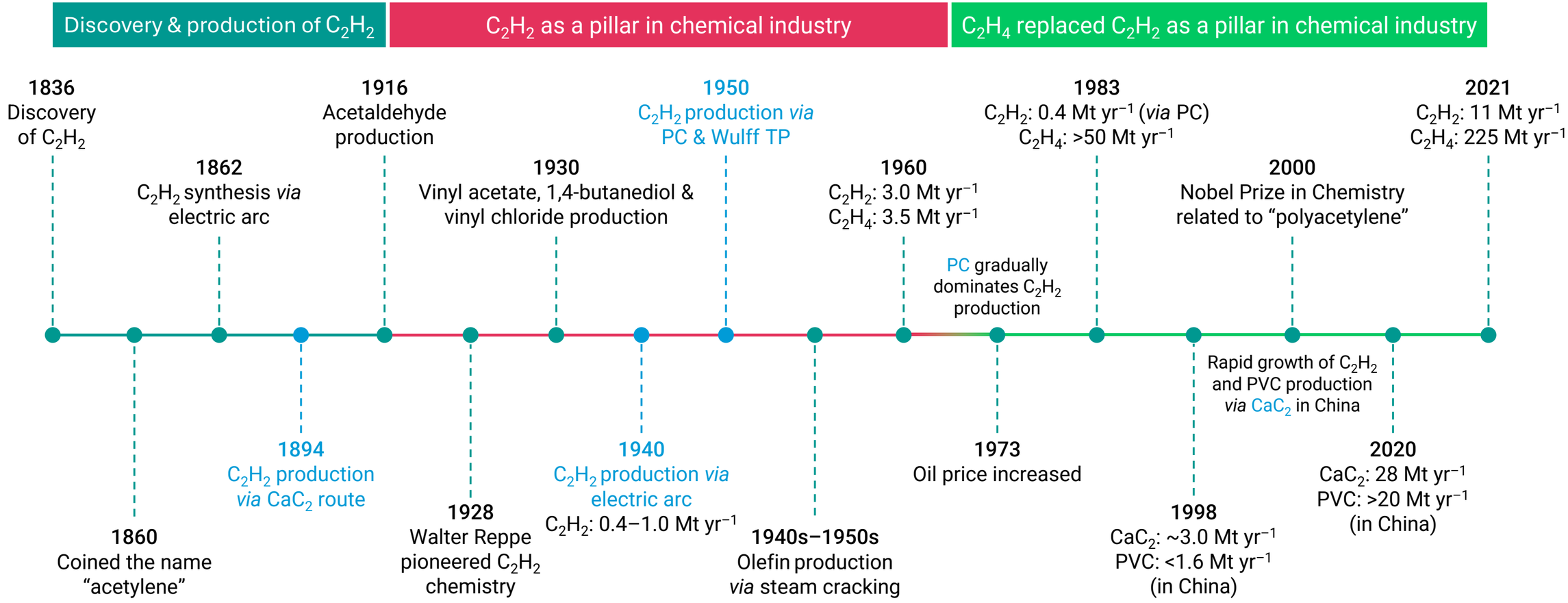
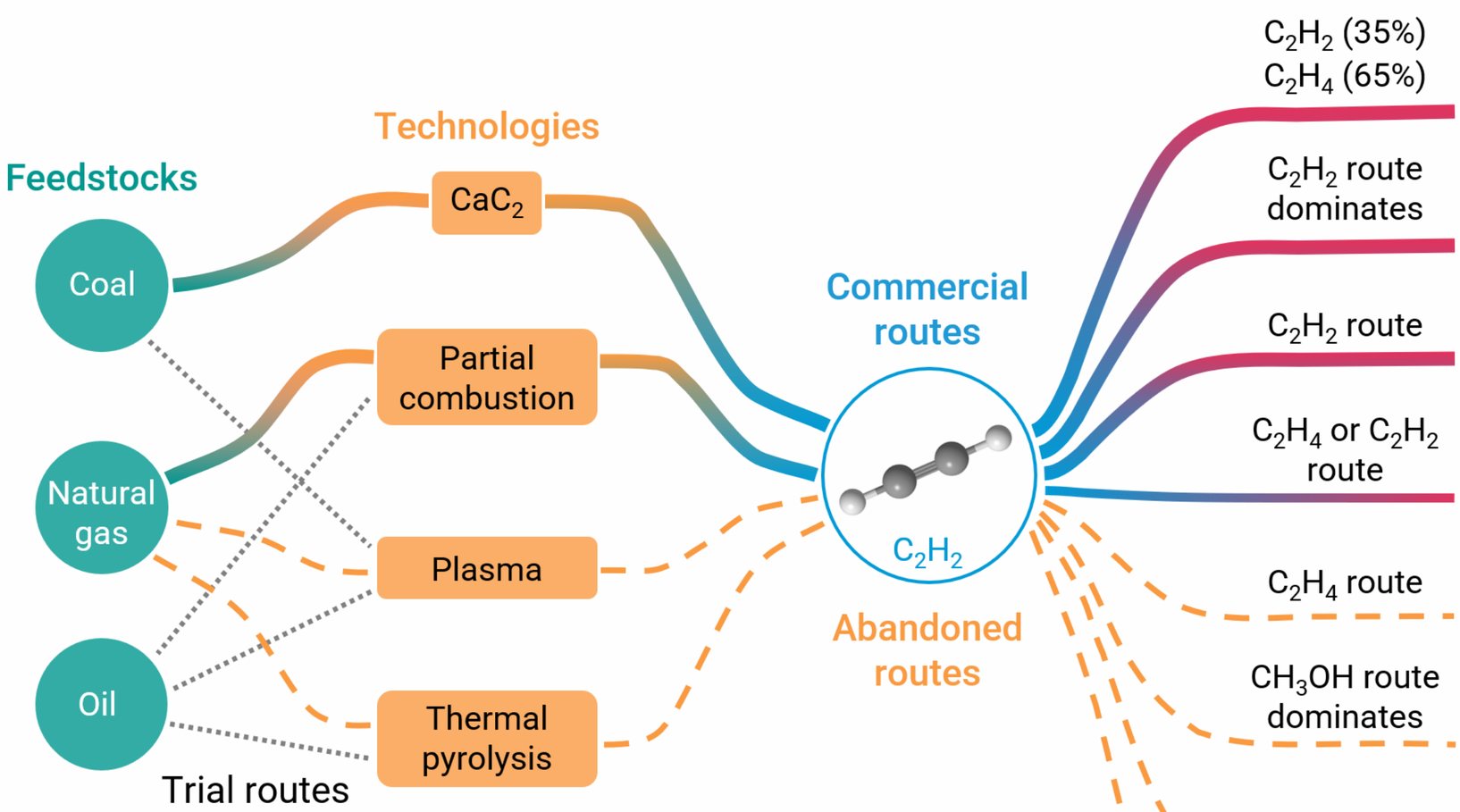
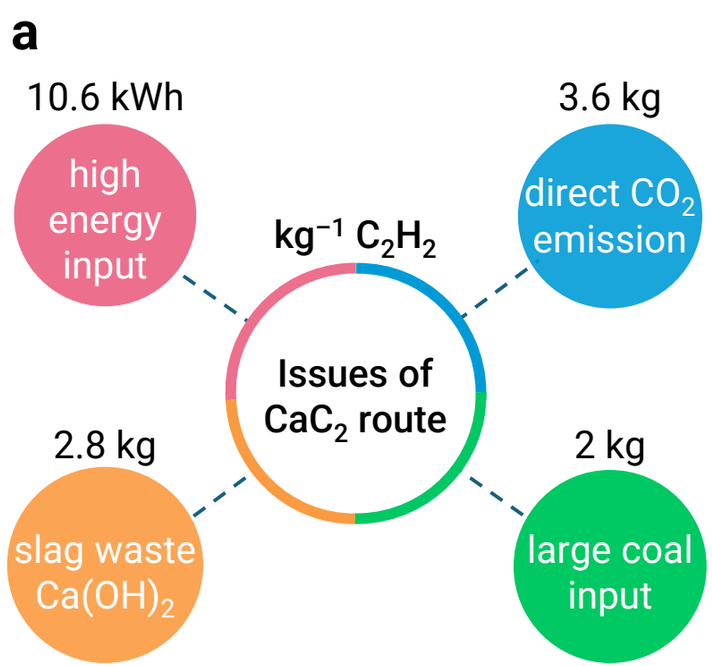
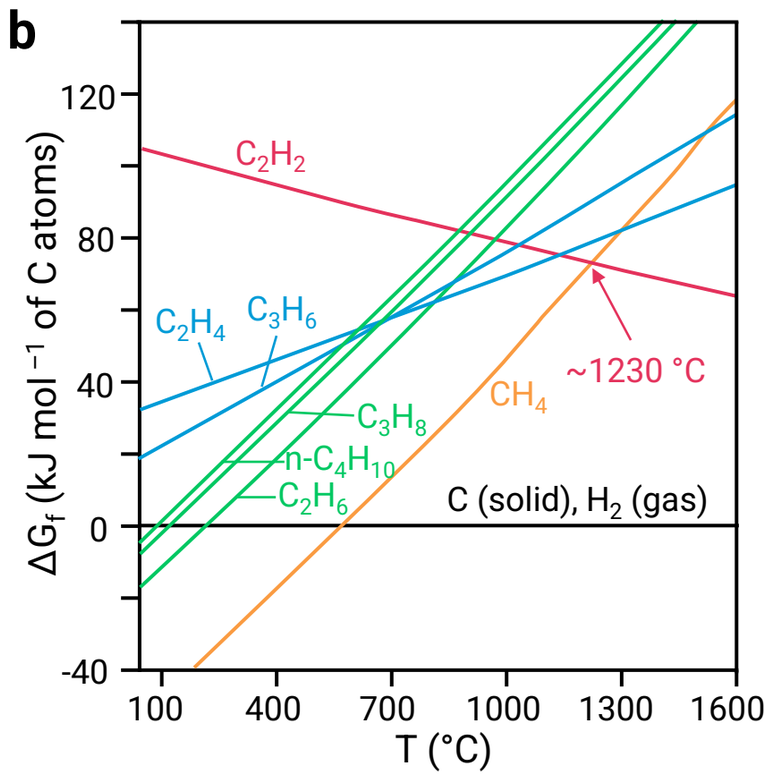

Discovery & production of C_2H_2
C_2H_2 as a pillar in chemical industry
C_2H_4 replaced C_2H_2 as a pillar in chemical industry
1836
Discovery of C_2H_2
1862
C_2H_2 synthesis *via* electric arc
1916
Acetaldehyde production
1930
Vinyl acetate, 1,4-butanediol & vinyl chloride production
1950
C_2H_2 production *via* PC & Wulff TP
1960
C_2H_2: 3.0 Mt yr^{-1}
C_2H_4: 3.5 Mt yr^{-1}
PC gradually dominates C_2H_2 production
1983
C_2H_2: 0.4 Mt yr^{-1} (*via* PC)
C_2H_4: >50 Mt yr^{-1}
2000
Nobel Prize in Chemistry related to "polyacetylene"
2021
C_2H_2: 11 Mt yr^{-1}
C_2H_4: 225 Mt yr^{-1}
1860
Coined the name "acetylene"
1894
C_2H_2 production *via* CaC_2 route
1928
Walter Reppe pioneered C_2H_2 chemistry
1940
C_2H_2 production *via* electric arc
C_2H_2: 0.4–1.0 Mt yr^{-1}
1940s–1950s
Olefin production *via* steam cracking
1973
Oil price increased
1998
CaC_2: ~3.0 Mt yr^{-1}
PVC: <1.6 Mt yr^{-1}
(in China)
Rapid growth of C_2H_2 and PVC production *via* CaC_2 in China
2020
CaC_2: 28 Mt yr^{-1}
PVC: >20 Mt yr^{-1}
(in China)



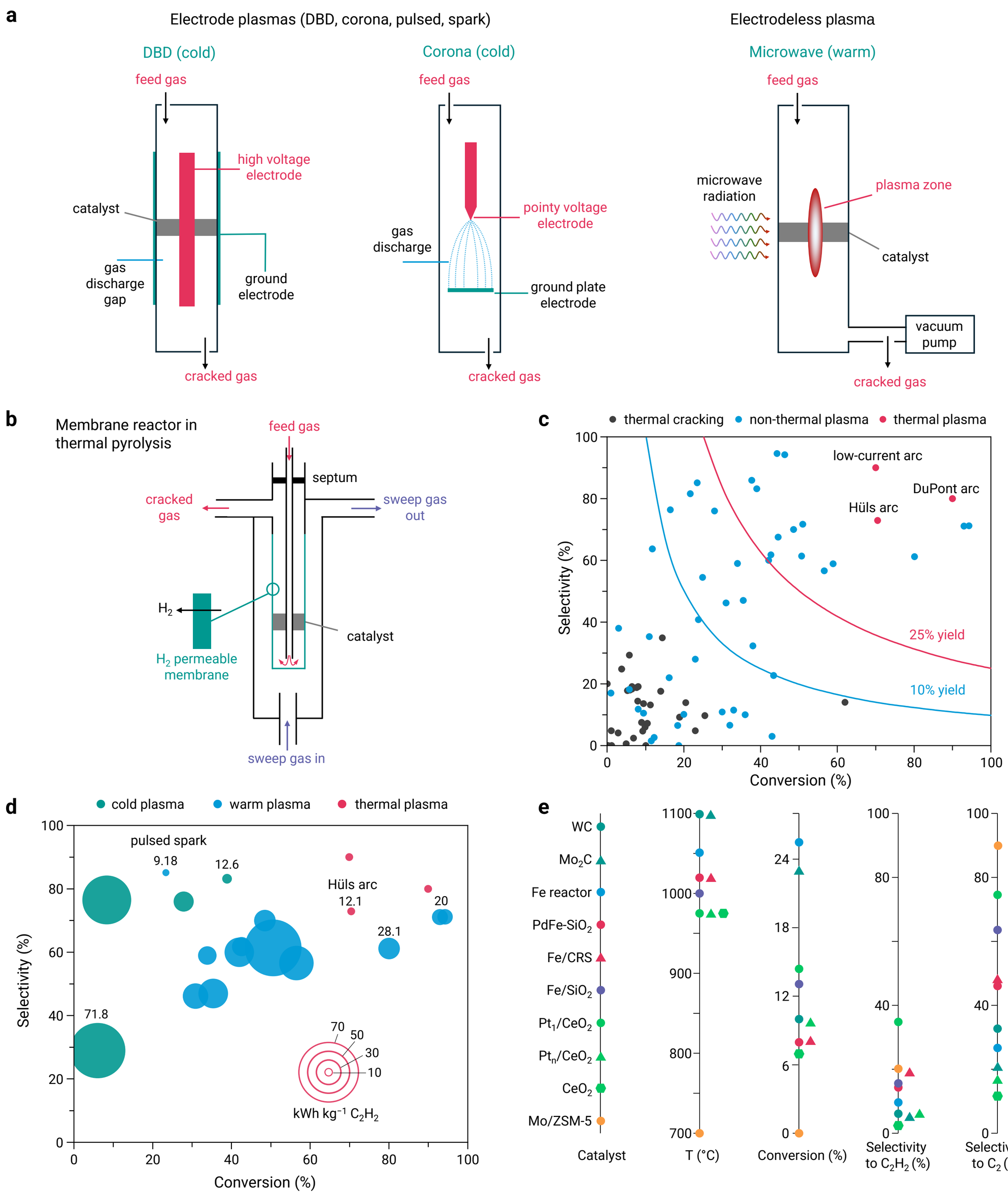
Commodity chemicals	Production (Mt yr^{-1})	Market size (USD billion yr^{-1})
vinyl chloride	50	68[a]
1,4-butanediol	2~2.5	6.5
acetylene black	0.1~0.2	0.3
vinyl acetate	6.2	9.0
acetaldehyde	1.5[b]	1.9[b]
acetic acid	18	15[b]
acrylic acid	8.8	23
acrylonitrile	7.0	12

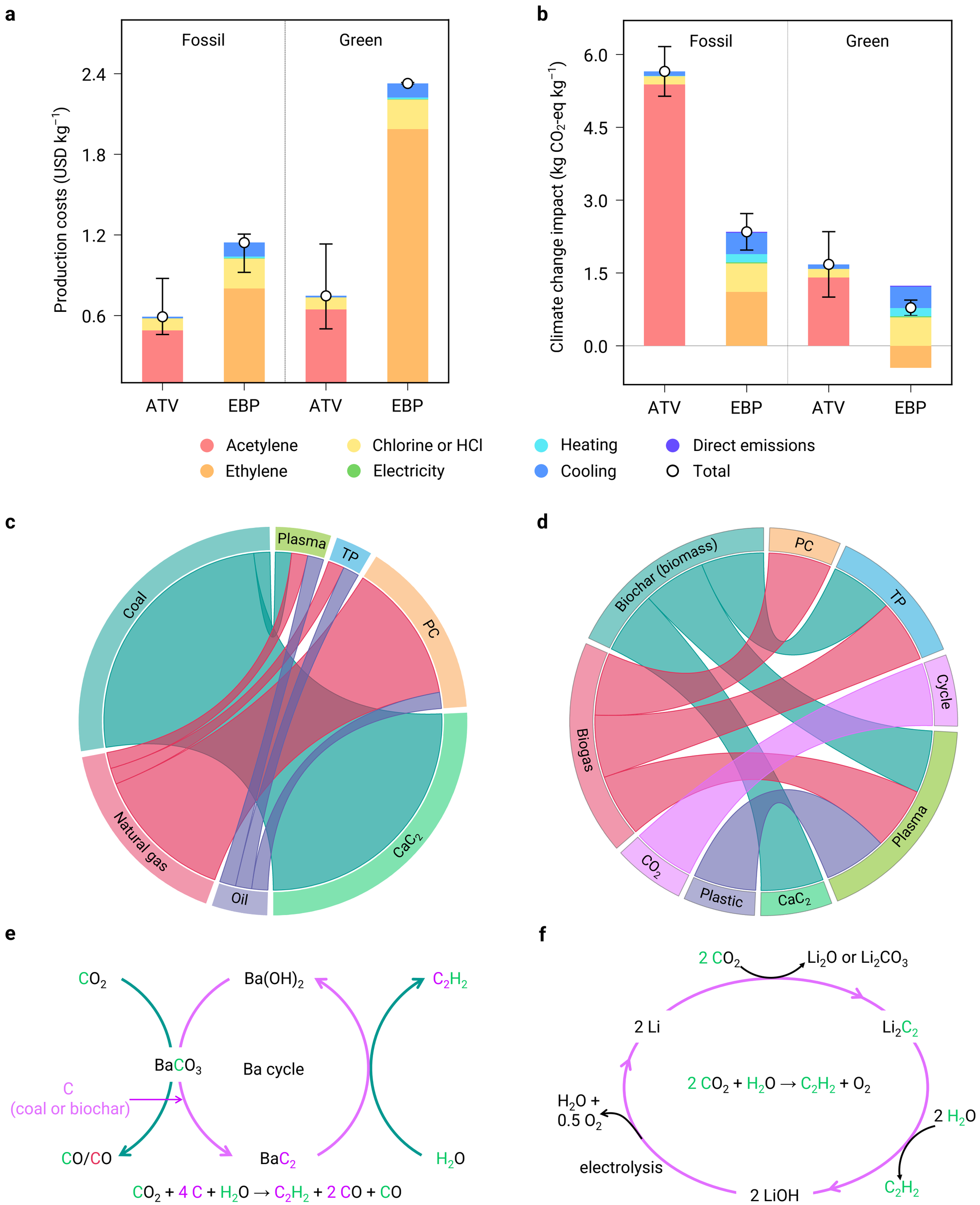


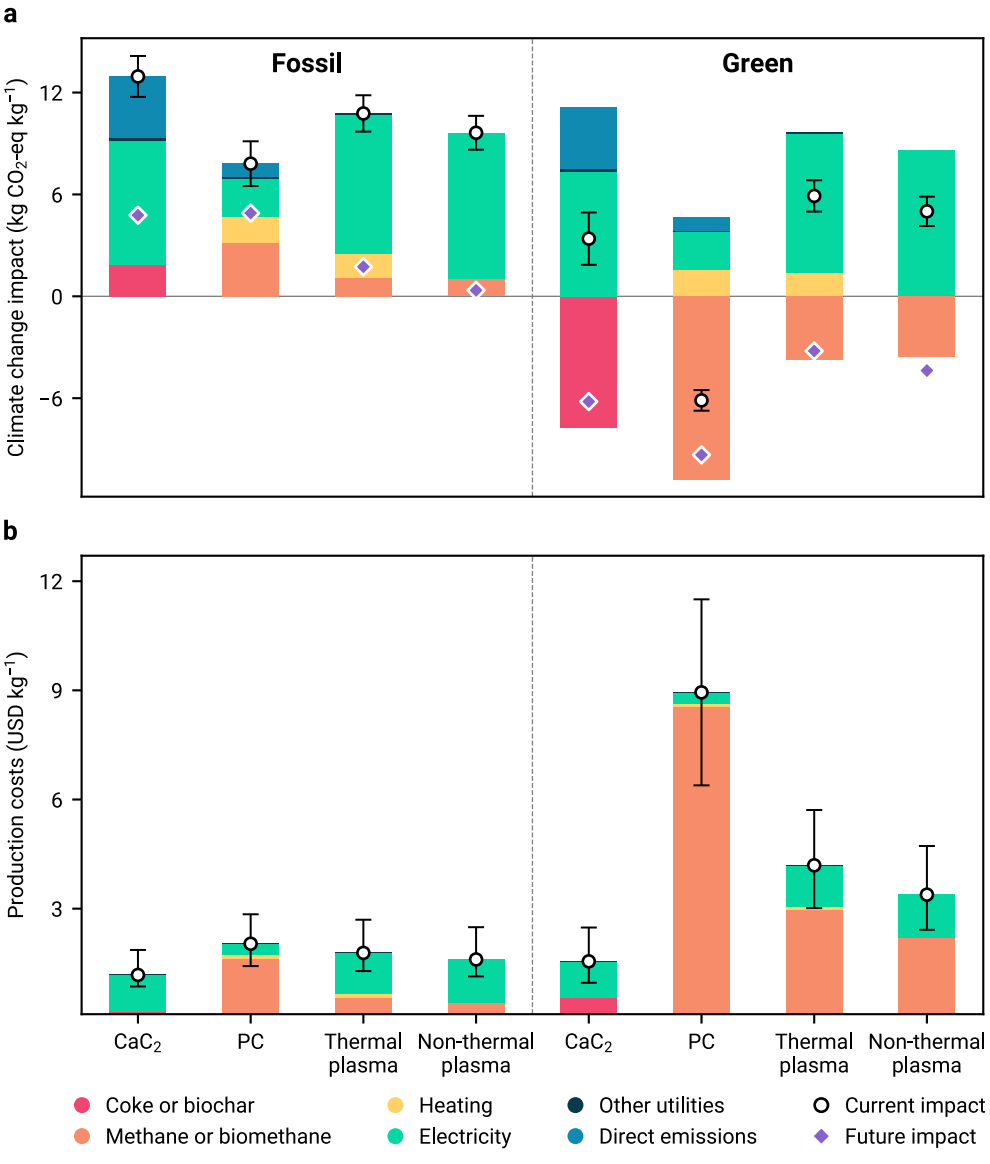


c Comparison of partial combustion and thermal plasma routes

C_2H_2 in cracked gas	8 vol%	11–15 vol%
energy input (kg^{-1} C_2H_2)	3.3 kWh	12.1 kWh
natural gas input (kg^{-1} C_2H_2)	5.7 Nm^3	2.0 Nm^3
CO_2 emission (kg^{-1} C_2H_2)	0.8 kg	~
soot formation (kg^{-1} C_2H_2)	50 mg	450 mg


a
Electrode plasmas (DBD, corona, pulsed, spark)
Electrodeless plasma
DBD (cold)
Corona (cold)
Microwave (warm)
feed gas
high voltage electrode
catalyst
gas discharge gap
ground electrode
cracked gas
pointy voltage electrode
gas discharge
ground plate electrode
microwave radiation
plasma zone
vacuum pump
b
Membrane reactor in thermal pyrolysis
septum
cracked gas
sweep gas out
H2
H2 permeable membrane
sweep gas in
c
thermal cracking
non-thermal plasma
thermal plasma
low-current arc
DuPont arc
Hüls arc
25% yield
10% yield
Selectivity (%)
Conversion (%)
d
cold plasma
warm plasma
thermal plasma
pulsed spark
9.18
12.6
Hüls arc
12.1
20
28.1
71.8
70
50
30
10
kWh kg-1 C2H2
Selectivity (%)
Conversion (%)
e
WC
Mo2C
Fe reactor
PdFe-SiO2
Fe/CRS
Fe/SiO2
Pt1/CeO2
Ptn/CeO2
CeO2
Mo/ZSM-5
Catalyst
T (°C)
Conversion (%)
Selectivity to C2H2 (%)
Selectivity to C2 (%)


a
Fossil
Green
Production costs (USD kg^{-1})
2.4
1.8
1.2
0.6
ATV
EBP
ATV
EBP
b
Fossil
Green
Climate change impact (kg CO_2-eq kg^{-1})
6.0
4.5
3.0
1.5
0.0
ATV
EBP
ATV
EBP
Acetylene
Ethylene
Chlorine or HCl
Electricity
Heating
Cooling
Direct emissions
Total
c
Plasma
TP
PC
Coal
Natural gas
Oil
CaC_2
d
PC
Biochar (biomass)
TP
Cycle
Biogas
Plasma
CO_2
Plastic
CaC_2
e
CO_2
$Ba(OH)_2$
C_2H_2
$BaCO_3$
Ba cycle
C
(coal or biochar)
CO/CO
BaC_2
H_2O
$CO_2 + 4\,C + H_2O \rightarrow C_2H_2 + 2\,CO + CO$
f
$2\,CO_2$
Li_2O or Li_2CO_3
2 Li
Li_2C_2
$2\,CO_2 + H_2O \rightarrow C_2H_2 + O_2$
H_2O +
$0.5\,O_2$
$2\,H_2O$
electrolysis
2 LiOH
C_2H_2


a
Fossil
Green
Climate change impact (kg CO_2-eq kg^{-1})
12
6
0
−6
b
Production costs (USD kg^{-1})
12
9
6
3
CaC_2
PC
Thermal plasma
Non-thermal plasma
CaC_2
PC
Thermal plasma
Non-thermal plasma
Coke or biochar
Methane or biomethane
Heating
Electricity
Other utilities
Direct emissions
Current impact
Future impact